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SO(14) Unification: A Machine-Verified Algebraic Scaffold with Phenomenological Predictions

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SO(14) is the smallest orthogonal group that contains both the Standard Model gauge group SO(10) and a gravity sector SO(4). Its 91 generators split as $45 + 6 + 40$: gauge, gravity, and the interactions between them. We build the algebraic scaffold for this theory and check every step with the Lean 4 theorem prover—87 proof files, over 2,900 verified declarations, zero `sorry` gaps.

What the computer checks: five certified Lie algebra homomorphisms composing $\mathfrak{su}(5) \hookrightarrow \mathfrak{so}(10) \hookrightarrow \mathfrak{so}(14) \hookrightarrow \mathfrak{so}(14)_M \hookrightarrow \mathfrak{so}(16)$, the Jacobi identity for nine Lie algebras including the 248-dimensional E_8, anomaly trace identities, and subalgebra closures for both SO(16) and SU(9) inside E_8. What the computer does not check: whether SO(14) describes nature.

The breaking chain runs $\mathrm{SO}(14) \to \mathrm{SO}(10)\times\mathrm{SO}(4) \to \cdots \to \mathrm{U}(1)_{\mathrm{EM}}$ via a symmetric traceless **104** Higgs. One-loop running [CO] gives coupling unification at $10^{16.98}$ GeV with a 0.88% miss, proton lifetime $10^{38.9}$ years (safe), and asymptotic freedom above the SM scale. Three open problems: three generations, Distler ghosts, and the mass gap. 87% of the algebraic results hold for all real forms of $\mathfrak{so}(14, \mathbb{C})$. No grand unified theory has previously received machine-checked algebraic scaffolding.

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I. INTRODUCTION

The Standard Model works. Three forces, 19 free parameters, no gravity. Grand unified theories try to fix this by embedding $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ in a bigger group [4–9]. SO(10) is the workhorse: one 16-dimensional spinor holds a complete generation of fermions [8]. But SO(10) says nothing about gravity.

SO(14) is the smallest orthogonal group that fits both SO(10) and the Lorentz group SO(1, 3) [13–15]. Nesti, Percacci, and Krasnov developed the physics. What was missing: a check that the algebra actually works.

That is what this paper provides. We built the algebraic scaffold in Lean 4 [37] with mathlib [38]: 87 proof files, over 2,900 verified declarations, zero `sorry` gaps. The proofs compile or they don't. Ours compile.

The scaffold is not just arithmetic. Five certified Lie algebra homomorphisms compose into a chain from $\mathfrak{su}(5)$ through $\mathfrak{so}(16)$. Nine Lie algebras carry verified Jacobi identities, including the 248-dimensional E_8 (3.78 billion evaluations via `native_decide`). Both SO(16) and SU(9) are verified as genuine subalgebras of E_8 via bracket closure. The anomaly trace identity $\mathrm{Tr}(A\{B, C\}) = 0$ is proved for antisymmetric matrices over any ring. These are algebraic theorems, not dimensional coincidences.

But there is also arithmetic. Dimension counts, Goldstone boson numbers, beta-coefficient inputs—all checked by `norm_num`. Both tiers are verified with the same deductive rigor as the Flyspeck project [39] or the liquid tensor experiment [40].

No grand unified theory has previously been machine-checked. From the outset, we distinguish what is proved from what is proposed:

- **Machine-verified** [MV]: Two tiers. *Structural* [MV-structural]: five certified Lie algebra homomorphisms ($\mathfrak{su}(5) \hookrightarrow \mathfrak{so}(10) \hookrightarrow \mathfrak{so}(14) \to \mathfrak{so}(14)_M \hookrightarrow \mathfrak{so}(16)$, plus $\mathfrak{so}(4) \hookrightarrow \mathfrak{so}(14)$), the Jacobi identity for nine Lie algebras (including the 248-dimensional E_8 via 3.78 billion evaluations), the A_4 Cartan matrix of $\mathfrak{su}(5) \subset \mathfrak{so}(10)$ (48 bracket relations), centralizer closure, gauge-gravity commutativity, anomaly trace identities, and $\mathfrak{so}(16) \subset E_8$ subalgebra closure. *Arithmetic* [MV-arithmetic]: dimensional consistency checks for anomaly prerequisites, Goldstone boson counts, beta-coefficient inputs, and representation dimensions. Both tiers hold for the complex Lie algebra $\mathfrak{so}(14, \mathbb{C})$ and all its real forms including SO(3, 11).

- **Computed** [CO]: Renormalization group running, coupling unification, proton lifetime, breaking chain physics.

- **Candidate physics** [CP]: The proposal that SO(14) (or SO(3, 11)) describes nature, the specific Higgs sector, the three-generation mechanism.

- **Open problems** [OP]: Three generations, Distler ghost objection, mass gap.

Machine verification does not prove that SO(14) describes nature. It proves that the algebra is consistent. Whether nature uses this algebra is a question for experiment.

Section II covers the machine-verified foundation. Section III builds the SO(14) model. Section IV runs the couplings. Section V extracts predictions. Section VI states the open problems honestly. Section VII says what we proved and what we didn't.

II. MACHINE-VERIFIED FOUNDATION

A. Lean 4 Proof Methodology

The algebraic scaffold is formalized in Lean 4 [37], a dependently-typed programming language and interactive theorem prover. Lean 4's type theory ensures that every theorem is deductively valid: the compiler rejects any proof containing logical gaps. The mathlib library [38] provides the mathematical infrastructure (real and complex numbers, linear algebra, ring theory).

Our proof strategy uses explicit finite-dimensional representations rather than mathlib's abstract `CliffordAlgebra` type. This choice prioritizes computational transparency: proofs reduce to `ext` (extensionality), `simp` (simplification), `ring` (polynomial identity), and `native_decide` (decidable computation), making them independently auditable. The complete methodology is described in the companion paper [41].

The codebase compiles as a single Lean 4 project (over 3,400 build jobs, zero errors, zero `sorry` gaps) and is available from the corresponding author upon reasonable request [44].

B. Key Machine-Verified Results

Table I summarizes the principal machine-verified theorems relevant to the SO(14) model.

C. Clifford Algebra Hierarchy

The algebraic path to SO(14) proceeds through a hierarchy of Clifford algebras [42]:

$$\mathrm{Cl}(1,1) \hookrightarrow \mathrm{Cl}(1,3) \hookrightarrow \mathrm{Cl}(6) \hookrightarrow \mathrm{Cl}(10) \hookrightarrow \mathrm{Cl}(14). \qquad (1)$$

At each level, the grade-2 elements form a Lie algebra under the commutator bracket (the "grade-2 Lie algebra functor"[MV]):

$$\mathrm{Cl}^2(n,0) \cong \mathfrak{so}(n), \qquad [A,B] = AB - BA. \qquad (2)$$

This construction produces the chain

$$\mathfrak{so}(1,1) \hookrightarrow \mathfrak{so}(1,3) \hookrightarrow \mathfrak{so}(6) \hookrightarrow \mathfrak{so}(10) \hookrightarrow \mathfrak{so}(14), \qquad (3)$$

where $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ (Pati-Salam), $\mathfrak{so}(10)$ (Georgi-Glashow-Fritzsch), and $\mathfrak{so}(14)$ (this work). Five certified Lie algebra homomorphisms compose into a chain: $\mathfrak{su}(5) \xrightarrow{\text{LieHom}} \mathfrak{so}(10) \xrightarrow{\text{LieHom}} \mathfrak{so}(14) \xrightarrow{\text{bridge}} \mathfrak{so}(14)_M \xrightarrow{\text{LieHom}} \mathfrak{so}(16)$, where $\mathfrak{so}(14)_M$ denotes the mathlib matrix representation (the "type bridge" connecting the flat algebraic construction to mathlib's `LieSubalgebra` infrastructure). Additionally, $\mathfrak{so}(4) \xrightarrow{\text{LieHom}} \mathfrak{so}(14)$ embeds the gravity sector. The E_8 Lie algebra (248 dimensions) is independently verified via sparse Jacobi identity with `native_decide`, and $\mathfrak{so}(16) \subset E_8$ is verified as a subalgebra closure.

D. Lie Algebra Identification: Beyond Dimension Counting

A common objection to machine-verified algebraic proofs is that they reduce to "just arithmetic"—proving dimensions add up without verifying genuine algebraic structure. The file `su5_lie_structure.lean` addresses this directly by proving that the 24-dimensional subalgebra of $\mathfrak{so}(10)$ identified in `su5_so10_embedding.lean` has the Lie algebra structure of type $A_4 = \mathfrak{su}(5)$.[MV]

The proof constructs explicit Cartan generators $H_1, \ldots, H_4$ and root generators R_{ab}, S_{ab} inside $\mathfrak{so}(10)$, then verifies all bracket relations. In the compact real form, each root space is 2-dimensional: the Cartan element H_i acts on the root space of α_j as

$$[H_i, R_{j,j+1}] = A_{ij} \cdot S_{j,j+1}, \quad [H_i, S_{j,j+1}] = -A_{ij} \cdot R_{j,j+1}, \qquad (4)$$

where A_{ij} is the Cartan matrix entry. The machine-verified eigenvalues exactly reproduce the A_4 Cartan matrix[MV]:

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}. \qquad (5)$$

In addition, co-root recovery ($[R_{j,j+1}, S_{j,j+1}] = 2H_j$) and root composition ($[R_{12}, R_{23}] = R_{13}$) are verified.[MV] By the Killing–Cartan classification theorem, the Cartan matrix uniquely determines the Lie algebra type [46]. The A_4 matrix corresponds to $\mathfrak{su}(5)$ and no other simple Lie algebra. This establishes the embedding as a genuine Lie algebra homomorphism of type A_4, not merely a dimensional coincidence. To our knowledge, this is the first machine-verified Lie algebra *identification* (not just dimension counting) for a GUT subalgebra.

E. What Machine Verification Does and Does Not Prove

Machine verification proves that the algebraic identities are *deductively valid*: if the axioms of Lean 4's type theory and the definitions in mathlib are consistent, then the theorems hold. This is a stronger guarantee than peer review, which checks plausibility rather than logical validity.

Machine verification does *not* prove:

1. That SO(14) describes nature (an empirical question).
2. That the full quantum field theory is well-defined (the mass gap problem[OP]).
3. That the non-compact form SO(3, 11) is ghost-free (the Distler objection[OP]).
4. That three generations arise naturally (open for all GUTs[OP]).

TABLE I. Principal machine-verified results. All proofs compile with zero `sorry` gaps in Lean 4 with mathlib. Results range from structural (Cartan matrix, Jacobi identity, centralizer closure) to arithmetic (dimension checks via `norm_num`).

Result	Lean File	Key Theorem	Content
$\dim \mathfrak{so}(14) = \binom{14}{2} = 91$	`so14_unification`	`so14_dimension`	Dimension counting
$91 = 45 + 6 + 40$	`so14_unification`	`unification_decomposition`	Dimension arithmetic (`norm_num`)
SU(5) ⊂ SO(10) via $[\mathfrak{g}, J] = 0$	`su5_so10_embedding`	`centralizer_closed`	Complex structure embedding
A_4 root system of $\mathfrak{su}(5)$	`su5_lie_structure`	`H1_R12`, `coroot_1`	Lie algebra identification
$\mathrm{Tr}(A\{B, C\}) = 0$ for antisymmetric	`anomaly_trace`	`trace_antisymm_anticommutator`	Algebraic anomaly identity
Anomaly prerequisites ($N \geq 7$, etc.)	`so14_anomalies`	`anomaly_checklist`	Dimensional checks for anomaly freedom
$Q = T_3 + Y/2$ preserves VEV	`symmetry_breaking`	`charge_preserves_vev`	Photon masslessness
$16 = 1 + 10 + \bar{5}$	`spinor_matter`	`spinor_decomposition`	One generation per spinor
Higgs: sym. traceless **104**	`so14_breaking_chain`	`so14_sym_traceless_dim`	Corrected Higgs rep
$75 + 16 = 91$	`so14_breaking_chain`	`gauge_boson_conservation`	Dimension arithmetic (`norm_num`)
$[\mathfrak{so}(4), \mathfrak{so}(10)] = 0$ in $\mathfrak{so}(14)$	`unification_gravity`	`gravity_gauge_commute`	Disjoint index sets
SM beta coefficient arithmetic	`rg_running`	`beta1_gut_normalized`	Arithmetic from assumed Dynkin indices
Jacobi identity for $\mathfrak{so}(10)$	`so10_grand`	`jacobi`	Lie algebra axiom
Jacobi identity for $\mathfrak{so}(14)$ (91 gens)	`so14_grand`	`jacobi`	Cubic polynomial, `ring`
E_8 Jacobi (248 dims, 3.78B evals)	`e8_sparse_jac_*`	`e8_jacobi_*`	Sparse `native_decide`
$\mathfrak{su}(5) \hookrightarrow \mathfrak{so}(10)$ LieHom	`su5c_so10_liehom`	`su5c_embed`	45 bracket equations
$\mathfrak{so}(10) \hookrightarrow \mathfrak{so}(14)$ LieHom	`so10_so14_liehom`	`so10_embed`	91 bracket equations
$\mathfrak{so}(14) \to \mathfrak{so}(14)_M$ type bridge	`so14_to_matrix`	`so14_bridge`	196 matrix entries
$\mathfrak{so}(14)_M \hookrightarrow \mathfrak{so}(16)_M$ LieHom	`so16_matrix`	`so14m_embed_block`	Block embedding
SU(9) ⊂ E_8 subalgebra closure	`e8_su9_subalgebra`	`su9_subalgebra_e8`	80 gens, `native_decide`
$\mathrm{Tr}(A\{B, C\}) = 0$ (antisymmetric)	`anomaly_trace`	`trace_anticomm_zero`	Anomaly trace identity
$\nabla \cdot B = 0$ from axiomatized Bianchi	`bianchi_identity`	`gauss_law_magnetism`	Consequence of axiomatized Bianchi identity

A signature audit of 46 of the 87 proof files [44] classifies 87% as *signature-independent* (properties of $\mathfrak{so}(14, \mathbb{C})$, holding for all real forms), 7% as *compact-only* (energy positivity results that fail for SO(3, 11) above the breaking scale), and 7% as *needs-revision* (results stated for a specific signature with natural Lorentzian analogues).

III. THE SO(14) MODEL

A. Algebraic Structure

The Lie algebra $\mathfrak{so}(14)$ has dimension $\binom{14}{2} = 91$[MV]. Under the subalgebra decomposition $\mathfrak{so}(14) \supset \mathfrak{so}(10) \oplus \mathfrak{so}(4)$, the 91 generators decompose as[MV]

$$91 = \underbrace{45}_{\text{gauge}} + \underbrace{6}_{\text{gravity}} + \underbrace{40}_{\text{mixed}}, \tag{6}$$

where:

- The 45 generators of $\mathfrak{so}(10)$ contain $\mathfrak{su}(5)$ (identified as type A_4 by its Cartan matrix[MV], Sec. II D) plus leptoquarks.
- The 6 generators of $\mathfrak{so}(4) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$ are identified with the Lorentz group in the non-compact form $\mathfrak{so}(3, 1)$.[CP]
- The $10 \times 4 = 40$ remaining generators mix gauge and gravitational indices; the representation identification as the $(\mathbf{10}, \mathbf{4})$ bifundamental follows from standard branching rules[SP].

The gauge and gravity subalgebras commute[MV]:

$$[\mathfrak{so}(10), \mathfrak{so}(4)] = 0 \quad \text{in } \mathfrak{so}(14), \tag{7}$$

which holds because the two subalgebras act on disjoint index sets (indices 1–10 vs. 11–14). The Kronecker deltas on disjoint indices vanish, forcing the bracket to zero. This result is signature-independent: it holds for $\mathfrak{so}(14, \mathbb{C})$ and all its real forms.[MV]

The SO(14) algebra is anomaly-free. For SO(N) with $N \geq 7$, all gauge anomalies vanish by standard group theory (the relevant trace invariants are identically zero). The machine verification checks the dimensional prerequisites for these conditions (e.g., $N = 14 \geq 7$); the vanishing itself follows from the representation theory of orthogonal groups[SP]. Anomaly freedom is a property of the complex Lie algebra and holds for all real forms, including SO(3, 11).

B. Breaking Chain

The symmetry-breaking chain from SO(14) to the Standard Model proceeds in five steps[CO] (individual steps verified in separate Lean files; a single end-to-end composition theorem is not yet formalized):

$$\begin{aligned} \mathrm{SO}(14) &\xrightarrow{\mathbf{104}} \mathrm{SO}(10) \times \mathrm{SO}(4) \\ &\xrightarrow{\mathbf{45}} \mathrm{SU}(5) \times \mathrm{U}(1)_\chi \times \mathrm{SO}(4) \\ &\xrightarrow{\mathbf{24}} \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y \times \mathrm{SO}(4) \\ &\xrightarrow{\mathbf{4}} \mathrm{SU}(3) \times \mathrm{U}(1)_{\mathrm{EM}} \times \mathrm{SO}(4). \end{aligned} \tag{8}$$

a. Step 1: SO(14) $\to$ SO(10) $\times$ SO(4). The Higgs field is a symmetric traceless 2-tensor S_{ab} (dimension $14 \times 15/2 - 1 = 104$)[MV], not the adjoint (dimension 91). This distinction is critical: an adjoint VEV breaks SO($2n$) $\to$ U(n), not to SO($2n_1$)$\times$SO($2n_2$) [28, 29]. The symmetric traceless VEV $\langle S \rangle = \mathrm{diag}(a, \ldots, a, b, \ldots, b)$ with $10a + 4b = 0$ (tracelessness) preserves exactly SO(10) $\times$ SO(4).[MV]

The 104 decomposes under SO(10) $\times$ SO(4) as[MV]

$$\mathbf{104} = (\mathbf{54}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{9}) \oplus (\mathbf{10}, \mathbf{4}) \oplus (\mathbf{1}, \mathbf{1}). \tag{9}$$

The $(\mathbf{10}, \mathbf{4})$ components (dimension $10 \times 4 = 40$, verified by arithmetic[MV]) are eaten as Goldstone bosons by the 40 broken gauge bosons, by the Goldstone theorem[SP]. The remaining $104 - 40 = 64$ components become physical massive scalars.

The Higgs potential for the symmetric traceless representation is

$$V(S) = -\mu^2 \mathrm{Tr}(S^2) + \kappa \mathrm{Tr}(S^3) + \lambda_1 [\mathrm{Tr}(S^2)]^2 + \lambda_2 \mathrm{Tr}(S^4), \tag{10}$$

where the cubic invariant $\kappa \neq 0$ is responsible for the SO(10)$\times$SO(4) vacuum alignment (as opposed to SO(7)$\times$ SO(7) or other patterns).

b. Steps 2–4. The remaining steps are standard GUT technology [7, 8]: SO(10) $\to$ SU(5) $\times$ U(1)$_\chi$ via an adjoint **45** VEV[SP] (dimension dim $\mathbf{45} = 45$[MV]); SU(5) $\to$ SU(3) $\times$ SU(2) $\times$ U(1)$_Y$ via an adjoint **24** VEV[SP] (dimension dim $\mathbf{24} = 24$[MV]); electroweak breaking via a fundamental Higgs doublet[SP] (dimension dim $\mathbf{4} = 4$[MV]).

c. Mass spectrum. The total number of broken generators across all steps is $40+20+12+3 = 75$[MV], producing 75 massive gauge bosons. The remaining $91-75 = 16$ generators are: 8 gluons, 1 photon, 6 SO(4) generators (gravity sector), and 1 U(1)$_\chi$.[MV] The U(1)$_\chi$ factor must be broken by an additional scalar (e.g., a spinor **16** of SO(10), contained in the **64** of SO(14)) to avoid a massless gauge boson excluded by experiment.[CP]

Table II summarizes the breaking chain with Goldstone boson counting.

TABLE II. Symmetry-breaking chain. "Broken" = number of broken generators = Goldstone bosons eaten. "Physical" = remaining scalar components. Dimension arithmetic is machine-verified [MV]; representation identifications follow from standard branching rules [SP].

Step	**Higgs**	**dim**	**Broken**	**Physical**
SO(14) $\to$ SO(10)$\times$SO(4)	**104**	104	40	64
SO(10) $\to$ SU(5)$\times$U(1)	**45**	45	20	25
SU(5) $\to$ SM	**24**	24	12	12
SU(2) $\times$ U(1) $\to$ U(1)$_{\mathrm{EM}}$	**4**	4	3	1
Total		177	75	102

C. Matter Content and Generations

The semi-spinor of Spin(14) has dimension $2^{14/2-1} = 2^6 = 64$. Under SO(14) $\to$ SO(10) $\times$ SO(4), where SO(4) $\cong$ SU(2)$_a$ $\times$ SU(2)$_b$, the semi-spinor dimension $2^6 = 64$ is machine-verified[MV]; the branching rule is standard [30][SP]:

$$\mathbf{64}_+ = (\mathbf{16}, (\mathbf{2}, \mathbf{1})) \oplus (\overline{\mathbf{16}}, (\mathbf{1}, \mathbf{2})), \tag{11}$$

$$\mathbf{64}_- = (\overline{\mathbf{16}}, (\mathbf{2}, \mathbf{1})) \oplus (\mathbf{16}, (\mathbf{1}, \mathbf{2})). \tag{12}$$

Each semi-spinor contains exactly one **16** of SO(10), which is exactly one generation of Standard Model fermions (including the right-handed neutrino)[CO].[SP] The $\overline{\mathbf{16}}$ is the conjugate representation (antimatter/mirror fermions), and the SU(2) doublet structure reflects the extra SO(4) degrees of freedom.

The full matter content is taken to be three copies of the semi-spinor[CP]:

$$\text{Matter} = 3 \times \mathbf{64}_+ \to 3 \times (\mathbf{16}, (\mathbf{2}, \mathbf{1})) \oplus 3 \times (\overline{\mathbf{16}}, (\mathbf{1}, \mathbf{2})). \tag{13}$$

a. The three-generation problem. The number three is put in by hand, just as in SO(10) with $3 \times \mathbf{16}$[OP]. This is the universal GUT generation problem: no known simple Lie group predicts exactly three generations from representation theory alone. The historical SO(14)-specific literature (Ida, Kayama, and Kitazoe [17]; Ma, Du, Xue, and Zhou [18]) found *four* generations from the full 128-dimensional Dirac spinor, not three. The Wilczek-Zee critique [19] showed that SO(18) family unification fails (8 + 8 mirror families); for SO(14), this critique is much less severe because a single semi-spinor gives only one family.

The SO(4) = SU(2)$_L$ $\times$ SU(2)$_R$ factor that appears in the decomposition SO(14) $\to$ SO(10) $\times$ SO(4) initially suggests a four-sector family structure, since the semi-spinor branches as $\mathbf{64}_+ = (\mathbf{16}, (\mathbf{2}, \mathbf{1})) \oplus (\overline{\mathbf{16}}, (\mathbf{1}, \mathbf{2}))$[SP]. However, standard representation theory shows that no mechanism internal to SO(14) can produce a $3+1$ splitting of these four sectors[CO]:

- **Lie algebra impossibility**[CO]: any mass operator from $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ acting on the family space $(\mathbf{2}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2})$ has eigenvalues $\{a+b, a-b, -a+b, -a-b\}$, which satisfy $m_1+m_4 = m_2 + m_3 = 0$. Three equal eigenvalues require $a = b = 0$.

- **Representation mismatch**[CO]: the family space carries the spinorial representation $(\mathbf{2}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2})$, not the vectorial $(\mathbf{2}, \mathbf{2})$. Under any diagonal SU(2) $\subset$ SO(4), the family space decomposes as $\mathbf{2} \oplus \mathbf{2}$ (two doublets), never as $\mathbf{3} \oplus \mathbf{1}$ (triplet + singlet). This excludes discrete flavor symmetries $A_4 \subset \mathrm{SO}(3)_{\mathrm{diag}}$.

- **Matter-mirror mixing**: of the four sectors, two carry the **16** (matter) and two the $\overline{\mathbf{16}}$ (mirror), so no relabeling produces three identical generations.

The generation mechanism must therefore be external. The most developed option is orbifold family unification [22]: SO($2N$) gauge theory on $M^4 \times S^1/(\mathbb{Z}_2 \times \mathbb{Z}_2')$ yields three chiral families with specific boundary conditions. This mechanism has been demonstrated for the SO($2N$) series [26] and applies directly to SO(14). Other external approaches include Calabi-Yau compactification via the SO(14) × U(1) ⊂ E_8 embedding with Euler characteristic $\chi = \pm 6$ [27], and the BenTov-Zee topological mechanism [25].

We emphasize that SO(14) neither solves nor worsens the three-generation problem relative to SO(10). It inherits the same open question. The impossibility argument above—showing that the SO(4) family structure cannot produce $3+1$ from any Lie algebra element[CO]—constrains future model building.

D. Signature and Gravity

For algebraic proofs (dimensions, embeddings, anomalies), we work with compact SO(14) or equivalently the complex Lie algebra $\mathfrak{so}(14, \mathbb{C})$. These results are signature-independent[MV].

For physics, the relevant real form is SO(3, 11) (or equivalently Spin(3, 11)), which contains SO(1, 3) × SO(10) as a block-diagonal subgroup—the Lorentz group acting on indices $\{1, 2, 3, 4\}$ and the GUT gauge group on indices $\{5, \ldots, 14\}$[CP]. Following Nesti and Percacci [13, 14], the gravitational degrees of freedom are identified with the SO(3, 1) sector of SO(3, 11), and the metric arises from the breaking pattern.

The signature distinction has physical consequences:

- In SO(14, 0) (compact): the Killing form is negative definite, ensuring energy positivity for Yang-Mills theory.[MV]
- In SO(3, 11) (non-compact): the Killing form is indefinite on the 40 mixed generators, producing negative-norm states (ghosts) in the unbroken phase above the symmetry-breaking scale[OP].

This is Distler and Garibaldi's ghost objection [21], originally directed at Lisi's E_8 proposal [20] but applicable to any non-compact gauge-gravity unification. We acknowledge this as the principal open problem of the model[OP].

Two approaches to resolving the ghost problem have been proposed:

1. **Percacci's broken-phase argument** [14]: below the symmetry-breaking scale, the effective theory has compact gauge group, and the 40 mixed generators acquire masses of order M_{GUT}. The ghosts decouple at low energies.
2. **Krasnov's chiral formulation** [15]: a reformulation using self-dual and anti-self-dual projections that avoids the indefinite-metric sector entirely.

Neither resolution is universally accepted. Our machine-verified results for compact signature apply to the effective low-energy theory below the breaking scale, where the gauge group *is* compact.

a. Spinor reality. The Clifford algebra isomorphism classes differ between signatures: $\text{Cl}(14, 0) \cong M(128, \mathbb{R})$ has complex spinors (Weyl but no Majorana-Weyl), while $\text{Cl}(11, 3) \cong M(128, \mathbb{R})$ has real spinors (Majorana-Weyl exists)[SP]. The existence of Majorana-Weyl spinors in SO(3, 11) is a feature, not a bug: it permits Majorana neutrino masses.

b. Signature audit. Of the 46 proof files audited for signature dependence: 40 files (87%) are signature-independent, 3 files (7%) are compact-only (energy positivity, Hilbert space positivity, mass gap), and 3 files (7%) need revision for specific-signature results[CO]. The precise classification is documented in Ref. [44].

IV. RENORMALIZATION GROUP ANALYSIS

A. Beta Functions at Each Scale

The one-loop beta function for a simple gauge group G with gauge coupling g is[SP]

$$\mu \frac{d\alpha_i^{-1}}{d \ln \mu} = -\frac{b_i}{2\pi}, \tag{14}$$

where $\alpha_i = g_i^2/(4\pi)$ and

$$b_i = -\frac{11}{3} C_2(G) + \frac{2}{3} \sum_f T(R_f) + \frac{1}{3} \sum_{\text{cs}} T(R_{\text{cs}}) + \frac{1}{6} \sum_{\text{rs}} T(R_{\text{rs}}), \tag{15}$$

with sums over Weyl fermions, complex scalars, and real scalars respectively.[SP] The Dynkin indices use the convention $T(\text{fund SO}(N)) = 1$, $T(\text{fund SU}(N)) = 1/2$.[SP]

The Standard Model beta coefficients with GUT normalization ($\alpha_1^{\text{GUT}} = \frac{5}{3}\alpha_Y$) are computed from assumed Dynkin indices and group-theory factors; the arithmetic combining these inputs is machine-verified in `rg_running.lean`[MV]:

$$b_3 = -7, \quad b_2 = -\frac{19}{6}, \quad b_1 = \frac{123}{50}. \tag{16}$$

Table III gives the beta coefficients at each scale of the breaking chain.

a. Field content. At each scale, the beta coefficients are determined by the particle content:

- **SM** (M_Z to M_4): Standard Model with right-handed neutrino.[SP]
- **SU(5) × U(1)**$_\chi$ (M_4 to M_3): 3 generations of $(\mathbf{10} + \overline{\mathbf{5}} + \mathbf{1})$ Weyl fermions, adjoint Higgs Σ' ($\mathbf{24}$, real), fundamental Higgs H_5 ($\mathbf{5}$, complex).[CO]

- **SO(10) × SO(4)** (M_3 to M_1): 3 generations of $[(\mathbf{16}, (\mathbf{2}, \mathbf{1})) + (\overline{\mathbf{16}}, (\mathbf{1}, \mathbf{2}))]$, SO(10) adjoint Higgs Σ ($\mathbf{45}$, real), remnant $(\mathbf{54}, \mathbf{1})$ scalars from $\mathbf{104}$.[CO]

- **SO(14)** ($> M_1$): $3 \times \mathbf{64}_+$ semi-spinor fermions, symmetric traceless Higgs S ($\mathbf{104}$, real). $C_2(\text{adj}) = 24$, $T_f = 24$ (3 spinors), $T_{\text{rs}}(\mathbf{104}) = 16$.[CO] The resulting $b_{\text{SO}(14)} = -208/3 < 0$ indicates asymptotic freedom[CO]

The Dynkin indices used are summarized in Table IV.

B. Coupling Evolution and Unification

The one-loop evolution equation is[SP]

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(\mu_0) - \frac{b_i}{2\pi} \ln \frac{\mu}{\mu_0}. \tag{17}$$

Using the PDG values $\alpha_{\text{EM}}^{-1}(M_Z) = 127.952$, $\sin^2\theta_W(M_Z) = 0.23122$, and $\alpha_s(M_Z) = 0.1180$[SP] [36], the SM couplings at M_Z are:

$$\alpha_1^{-1} = 59.00, \quad \alpha_2^{-1} = 29.57, \quad \alpha_3^{-1} = 8.47. \tag{18}$$

a. Desert hypothesis. With a single unification scale (the α_2–α_3 crossing), we find[CO]:

$$M_{\text{GUT}} = 10^{16.98}\,\text{GeV}, \quad \alpha_{\text{GUT}}^{-1} = 47.03. \tag{19}$$

The α_1 coupling at M_{GUT} is $\alpha_1^{-1} = 45.46$, giving an absolute miss of 1.57 units of α^{-1} (3.3% relative miss). This is the well-known non-supersymmetric GUT triangle [11], identical to what SO(10) or SU(5) predict in the desert scenario.

b. Multi-scale approach. The SO(14) breaking chain introduces additional thresholds. With three free scales ($M_4 = 10^{16.50}\,\text{GeV}$, $M_3 = 10^{17.00}\,\text{GeV}$, $M_1 = 10^{19.00}\,\text{GeV}$), the best-fit coupling spread at M_1 is[CO]

$$\frac{\text{spread}}{\text{average}} = 0.88\%, \tag{20}$$

well below any reasonable threshold for failure.

TABLE III. One-loop beta coefficients at each scale. The SM coefficients are machine-verified [MV]; higher-scale coefficients are computed [CO]. Negative b indicates asymptotic freedom.

Scale	Group	b	AF?
M_Z to M_4	SU(3)	-7	Yes
M_Z to M_4	SU(2)	$-19/6$	Yes
M_Z to M_4	U(1)$_Y$	$+123/50$	No
M_4 to M_3	SU(5)	$-40/3$	Yes
M_3 to M_1	SO(10)	-38	Yes
M_3 to M_1	SU(2)$_a$	$+29/3$	No
$> M_1$	SO(14)	$-208/3$	Yes

The 0.88% residual miss can plausibly be closed by threshold corrections from the rich Higgs sector ($\mathbf{104} + \mathbf{45} + \mathbf{24} + \mathbf{4} = 177$ real scalar components), two-loop corrections, or split multiplet spectra[CO]. This degree of unification approaches the MSSM prediction of near-exact unification [12], though our model is non-supersymmetric.

C. Asymptotic Freedom

At all scales above the Standard Model, the gauge couplings are asymptotically free (except for U(1)$_Y$ and SU(2)$_a$ of the SO(4) sector). The SO(14) beta coefficient $b = -208/3 \approx -69.3$ is large and negative[CO], indicating strong asymptotic freedom. The SO(4) sector's SU(2)$_a$ factor has $b = +29/3 > 0$, meaning it is not asymptotically free; however, this factor is entirely decoupled from the Standard Model coupling evolution (it consists of SM singlets)[CO].

V. PHENOMENOLOGICAL PREDICTIONS

A. Proton Decay

The dominant proton decay channel in any GUT containing SU(5) as a subgroup is mediated by the X and Y leptoquark gauge bosons, with mass $M_X \sim M_4$.[SP] The proton lifetime scales as[SP]

$$\tau_p \sim \frac{M_X^4}{\alpha_{\text{GUT}}^2 m_p^5} \cdot \frac{1}{A_R^2}, \tag{21}$$

where $A_R \approx 2.5$ is the short-distance renormalization enhancement factor[SP] [33].

With $M_X = M_{\text{GUT}} = 10^{16.98}\,\text{GeV}$ and $\alpha_{\text{GUT}} = 1/47.03$[CO]:

$$\tau_p \approx 10^{38.9}\ \text{years}. \tag{22}$$

This exceeds the current Super-Kamiokande bound of $\tau_p > 10^{34.4}$ years [34] by a factor of $\sim 3 \times 10^4$[CO].

TABLE IV. Dynkin indices for representations relevant to the breaking chain. Convention: $T(\text{fund SO}(N)) = 1$, $T(\text{fund SU}(N)) = 1/2$.

Group	Rep	dim	$T(R)$
SO(14)	adjoint **91**	91	24
SO(14)	spinor **64**	64	8
SO(14)	sym. traceless **104**	104	16
SO(10)	adjoint **45**	45	16
SO(10)	spinor **16**	16	2
SO(10)	sym. traceless **54**	54	12
SU(5)	adjoint **24**	24	5
SU(5)	antisymmetric **10**	10	3/2

The proton lifetime is safely above the experimental bound and will remain so even with the next-generation Hyper-Kamiokande experiment (projected sensitivity $\sim 10^{35.3}$ years [35]).

The proton lifetime in SO(14) is *identical* to that in minimal SO(10), because it depends only on M_4 and $\alpha_{\rm GUT}$, both of which are determined by the Standard Model couplings at M_Z and the SM beta coefficients[CO]. The SO(14)-specific gauge bosons (40 mixed generators) live at $M_1 \gg M_4$ and contribute to proton decay only as $(M_4/M_1)^4 \sim 10^{-10}$ corrections.

B. Novel SO(14) Signatures

Beyond the standard GUT predictions shared with SO(10), SO(14) introduces three classes of novel signatures[CP]:

a. 1. The 40 mixed gauge bosons. The $(\mathbf{10}, \mathbf{4})$ bifundamental gauge bosons, with masses $\sim M_1 \approx 10^{19}$ GeV, mediate transitions between gauge and gravitational indices. Their exchange produces dimension-6 operators coupling Standard Model fermions to the gravity sector. At low energies, these operators are suppressed by $(M_Z/M_1)^2 \sim 10^{-34}$, rendering them unobservable with current technology[CO].

b. 2. The SO(4) sector. The $\mathrm{SO}(4) = \mathrm{SU}(2)_a \times \mathrm{SU}(2)_b$ sector contains 6 generators that are Standard Model singlets. If the SO(4) symmetry is broken at a scale Λ_G below M_1, the resulting massive spin-1 bosons could have observable consequences. The nature of these consequences depends on the breaking scale and pattern, which are not determined by the algebraic structure alone[CP].

c. 3. The enhanced scalar sector. The 177 total scalar components $(104 + 45 + 24 + 4)$ provide a much richer threshold correction landscape than minimal SO(10) models. While the extra scalars are too heavy to be directly produced, their virtual effects on precision observables (the ρ parameter, flavor-changing neutral currents) could provide indirect signatures[CO].

C. Comparison with Other Unification Proposals

Table V compares SO(14) with other unification proposals.

The principal advantages of SO(14) relative to SO(10) are:

1. **Gravity inclusion**: the SO(3,1) Lorentz group is embedded as a subgroup, with $[\mathfrak{so}(10), \mathfrak{so}(4)] = 0$[MV] (proven for compact indices; the result is signature-independent and holds for $\mathfrak{so}(1,3)$ in the Lorentzian real form).

2. **Machine verification**: structural algebraic identities and dimensional prerequisites have been formally checked by computer.

3. **Four-sector spinor structure**: the SO(4) factor provides a four-sector decomposition $(\mathbf{2}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2})$ of the semi-spinor (dimension verified[MV]), with an impossibility argument constraining future generation mechanisms[CO]—structural information not available to SO(10).

The principal disadvantages are:

1. **Ghost problem**: the non-compact form SO(3,11) has indefinite Killing form on the 40 mixed generators[OP].

2. **Complexity**: 91 generators vs. 45, with additional breaking scales and scalar content.

3. **No improvement in unification**: the coupling unification quality is determined by the SM desert regime, which is identical[CO].

VI. DISCUSSION

A. What Is New Relative to SO(10)

The decomposition $91 = 45+6+40$ (dimensional arithmetic, verified by `norm_num`[MV]) reflects the subalgebra structure $\mathfrak{so}(10) \oplus \mathfrak{so}(4)$ inside $\mathfrak{so}(14)$. It shows that SO(14) is the minimal orthogonal group containing both the SO(10) GUT gauge group and a 6-dimensional sector SO(4) (the Lorentz group SO(3,1) in the non-compact form). The 40 mixed generators are genuinely new: they have no analogue in SO(10) and represent interactions that couple gauge and gravitational degrees of freedom.

This algebraic fact was noted by Nesti and Percacci [13, 14] and developed by Krasnov [15, 16]. Our contribution is to place the entire algebraic structure on machine-verified foundations and to compute the phenomenological consequences (coupling unification, proton decay, asymptotic freedom) with explicit claim tagging.

B. Open Problems

We state three principal open problems:

a. 1. Three generations[OP]. No simple Lie group is known to predict exactly three generations from representation theory alone. Standard representation-theoretic arguments show that no mechanism intrinsic to SO(14) can produce $3 + 1$ from the SO(4) family structure (Sec. III C): the Lie algebra impossibility argument, the representation mismatch (spinorial $2 + 2$ vs. vectorial $3 + 1$), and the matter-mirror content collectively exclude all internal candidates[CO]. More generally, the double-commutant theorem forces the family symmetry of *any* $\mathrm{SO}(10) \times \mathrm{SO}(4)$ decomposition to be exactly SO(4), whose spinorial representations have minimum dimension 2. Since $3 \nmid 2$, the $\mathbf{16}$ of SO(10) always appears

TABLE V. Comparison of unification proposals. "AF" = asymptotically free. "MV" = machine-verified algebraic structure. "Gravity" = gravity sector included in the gauge group. "Generations" = whether three generations arise naturally.

	$\dim G$	$\dim R_{\text{ferm}}$	AF?	MV?	Gravity?	Generations	τ_p (years)
SU(5) [7]	24	$\mathbf{\bar{5}}+\mathbf{10}$	Yes	No	No	3 by hand	10^{31} (excluded)
SO(10) [8]	45	**16**	Yes	No	No	3 by hand	10^{35}–10^{39}
E_6 [10]	78	**27**	Yes	No	No	3 by hand	model-dependent
SO(14) [this work]	**91**	**64**	**Yes**	**Yes**	**Yes**	**3 by hand**	$\mathbf{10^{38.9}}$
E_8 [20]	248	**248**	—	Partial*	Yes	claimed	not computed
SO(18) [19]	153	**256**	No	No	No	8+8 mirror	killed

with even multiplicity in any spinorial representation of SO(14)—a *spinor parity obstruction* that excludes all intrinsic mechanisms at once[CO]. The generation mechanism must therefore be external. The most developed option is orbifold family unification [22], demonstrated for the SO($2N$) series. Alternatively, Spin(3, 11) embeds in $E_{8(-24)}$ via $\mathrm{Spin}(3,11) \subset \mathrm{Spin}(12,4) \subset E_{8(-24)}$, and Wilson [23] has identified discrete order-3 elements of E_8 whose centralizer $\mathrm{SU}(9)/\mathbb{Z}_3$ provides a fundamentally discrete three-generation symmetry not visible at the SO(14) level. In a companion paper [43], we prove this E_8 resolution with full machine verification: the impossibility $3 \nmid 2$ in SO(14) and the resolution via $\Lambda^3(\mathbb{C}^9) = 84 \supset 3 \times 16 = 48$ in E_8 are verified across 11 Lean 4 files with zero `sorry` gaps. E_8 itself is now machine-verified as a Lie algebra (248 dimensions, 7,752 nonzero brackets, Jacobi identity verified on 3.78 billion evaluations via `native_decide`). Both maximal subgroups—SO(16) (120 generators) and SU(9) (80 generators)—are verified as genuine Lie subalgebras of E_8 via bracket closure[MV], and each J eigenspace of the $\mathbb{Z}_3$ automorphism is independently anomaly-free[MV]. The chirality question remains an open problem in mathematical physics[OP]: Distler–Garibaldi [21] proved that no embedding of the Standard Model in any real form of E_8 yields three *massless chiral* generations from the adjoint, but Wilson [24] argues that for theories with explicitly non-zero masses, this no-go theorem does not apply—the non-compact degrees of freedom serve as mass parameters rather than ghosts.

b. 2. Distler ghosts[OP]. The non-compact form SO(3, 11) has indefinite Killing form, producing negative-norm states for the 40 mixed generators [21]. Percacci's broken-phase argument [14] and Krasnov's chiral formulation [15] offer partial resolutions, but neither has been universally accepted. A rigorous proof of ghost decoupling or a reformulation of the theory that avoids ghosts entirely is needed. Our machine-verified energy positivity results (3 files, 7% of theorems) apply only to the compact form and do not resolve this question.

c. 3. Mass gap[OP]. The existence of a mass gap in four-dimensional Yang-Mills theory is the Clay Millennium Prize problem [45]. Our mass gap formalization axiomatizes the statement for the compact form, assuming energy positivity; the axioms are machine-checked but the mass gap itself is not proved (it is the Millennium Prize problem). The physically relevant non-compact theory requires a modified formulation. We make no claim on the mass gap beyond stating it precisely and checking the dimensional prerequisites.

C. Machine Verification as Methodology

Interactive theorem provers have not previously been applied to GUT physics. The main methodological advantages are:

1. **Error catching**: the symmetric traceless **104** (not adjoint **91**) correction for Step 1 was discovered during formalization, when Lean rejected the claim that adjoint VEVs break $\mathrm{SO}(2n) \to \mathrm{SO}(2n_1) \times \mathrm{SO}(2n_2)$.

2. **Completeness**: all 91 generators are defined, dimensional prerequisites for anomaly freedom are checked, Goldstone boson dimension counts are verified, and gauge boson mass spectrum accounting is confirmed. Structural results (Cartan matrix, Jacobi identity, centralizer closure) go beyond arithmetic.

3. **Reproducibility**: the codebase compiles deterministically. Any researcher with a Lean 4 installation can verify the results in minutes.

4. **Signature audit**: classifying proofs by signature dependence (87% independent, 7% compact-only, 7% needs-revision) provides a precise answer to the question "which results transfer to SO(3, 11)?"

We propose that machine verification become standard practice for new GUT proposals, just as gauge coupling unification and proton decay calculations are standard today. The upfront cost (formalization in Lean 4) is repaid by the elimination of algebraic errors that can persist undetected for decades in the literature.

D. Future Directions

Several concrete research directions emerge:

1. **Two-loop RG analysis**: the two-loop beta coefficients for SO(14) have not been computed. These could shift the unification scale and quality.

2. **Threshold corrections**: systematic computation of threshold corrections from the 177-scalar spectrum to determine whether the 0.88% coupling miss can be naturally closed.

3. **Orbifold three-generation construction**: explicit boundary conditions for SO(14) on $M^4 \times S^1/(\mathbb{Z}_2 \times \mathbb{Z}_2')$ following Kawamura-Miura [22], specializing their SO($2N$) framework to $N = 7$.

4. E_8 **generation mechanism** (partially complete): E_8 is now machine-verified as a Lie algebra (248 dimensions, Jacobi on 3.78 billion evaluations[MV]), $\mathrm{SO}(16) \subset E_8$ is verified as a subalgebra closure[MV], and the three-generation mechanism via $\mathrm{SU}(9)/\mathbb{Z}_3$ is dimensionally verified across 10 proof files [43]. Each J eigenspace of the $\mathbb{Z}_3$ automorphism is independently anomaly-free[MV]. Remaining: connecting the $\mathrm{Spin}(3, 11)$ real form to the verified compact scaffold, and resolving the chirality definition for massive fermions in $E_{8(-24)}$.

5. **Ghost resolution**: a rigorous treatment of ghost decoupling in the broken phase of $\mathrm{SO}(3, 11)$, potentially following Krasnov's chiral approach.

6. **Gravitational corrections**: at $M_1 \sim 10^{19}\,\mathrm{GeV}$ (near M_{Planck}), gravitational loop corrections to gauge couplings become relevant and could affect unification quality.

7. **Lean 4 formalization of the RG analysis**: extending the machine verification from algebra to the analytic results (evolution equations, matching conditions).

VII. CONCLUSION

Here is what we built. 87 Lean 4 proof files. Over 2,900 verified declarations. Zero `sorry` gaps. Five Lie algebra homomorphisms chain $\mathfrak{su}(5)$ through $\mathfrak{so}(16)$. E_8 is verified as a Lie algebra. Both SO(16) and SU(9) close as subalgebras inside it. The proofs compile. Anyone can check.

Here is what the proofs say about SO(14). The algebra is consistent: 91 generators split $45 + 6 + 40$, the embeddings preserve brackets, the anomaly traces vanish, the Goldstone counts work out. One-loop running [CO] gives $M_{\mathrm{GUT}} = 10^{16.98}\,\mathrm{GeV}$, proton lifetime $10^{38.9}$ years, and asymptotic freedom above the SM scale.

Here is what the proofs do not say. They do not say SO(14) describes nature. Three open problems remain: why three generations (every GUT has this problem), ghost states in the non-compact form (every gravity-gauge unification has this problem), and the mass gap (that is the Millennium Prize problem).

The algebra can't be wrong. The physics might be. We think that distinction is worth making precisely, and that machine verification should be standard practice for future GUT proposals.

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DATA AVAILABILITY

All data generated during this study are contained within the article. The Lean 4 source code for all machine-verified proofs is available from the corresponding author upon reasonable request [44].

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